



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

Dept. of Computer Science
Faculty of Science and Technology

CSC2210: OBJECT ORIENTED PROGRAMMING 2

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Section: M

Group No: 10

Project Report On Multi-Vendor E-Commerce System

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Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)		Total =
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described.	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive

	of verification processes, and data flow is not considered.	inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	ensuring data integrity and appropriate data flow within the system.	and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.
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CO3: Prepare and explain a real-life desktop-based application synthesizing several components of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	Evaluation Definition				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of databases ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identify and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanisms have been organized in a preferred way according to the coherent usage.

Table of Contents:	Page no.
1. Chapter :01 (Introduction)-----	04
2. Chapter :02(User Story) -----	05
3. Chapter :03 (Schema Diagram) -----	07
4. Chapter :04 (SQL Queries) -----	08
5. Chapter:05 (Screenshots) -----	11

Chapter 1: Introduction

The **Multi-Vendor E-Commerce System (ShopHub)** is a comprehensive desktop application developed using C#.NET Windows Forms and Microsoft SQL Server as the backend. Designed to facilitate decentralized online retail operations, the platform establishes an integrated digital marketplace where multiple independent merchants (Vendors) can register, manage, and sell their inventory, while buyers (Customers) can discover products, manage shopping carts, execute secure purchases, and rate items.

The application implements a robust, centralized administration hub where system administrators oversee user management, review merchant verification request lines, handle product categories, monitor platform transactions, and analyze business performance reports.

Key Features of the System:

- **Role-Based Access Control (RBAC):** Distinct administrative interfaces, business-management dashboards, and shopping portals tailored to three primary user classes: **Administrators**, **Vendors**, and **Customers**.
- **Customer Shopping Lifecycle:** Full checkout integration enabling product discovery (categorization, keyword queries), responsive in-memory cart operations, secure billing/shipping collection, stock level validation, and post-purchase item review systems.
- **Vendor Inventory & Storefront Management:** Real-time metrics overview, inventory CRUD capabilities, store detail personalization, and independent order fulfillment progression.
- **Security & Audit Trails:** Secure custom registration, password mutation layers, database-level reference integrity, and system-wide verification gates (such as Vendor storefront approvals).

Chapter 2: User Story

The system's features are defined by the following role-based user stories:

A. General / Anonymous User Stories

- **Sign Up:** As an anonymous user, I want to create an account as either a customer or a Vendor so that I can access the platform's role-specific features.
- **Login:** As a registered user, I want to securely log in to the system with my credentials so that I can be redirected to my respective dashboard.

B. Customer User Stories

- **Product Discovery:** As a customer, I want to browse products, filter them by category, and search by keywords so that I can quickly find the items I wish to buy.
- **Shopping Cart Management:** As a customer, I want to add products to my cart, update quantities, and remove items so that I can manage my order before checking out.
- **Checkout & Order Placement:** As a customer, I want to place an order by providing my payment method and shipping address so that I can purchase my selected products.
- **Order History:** As a customer, I want to view my past orders, check their statuses, and inspect individual order details so that I can track my purchases.
- **Product Reviews:** As a customer, I want to leave a rating and review for products I have purchased so that I can share my feedback with other shoppers.
- **Profile Personalization:** As a customer, I want to update my personal profile details and change my password so that I can keep my account details secure.

C. Vendor User Stories

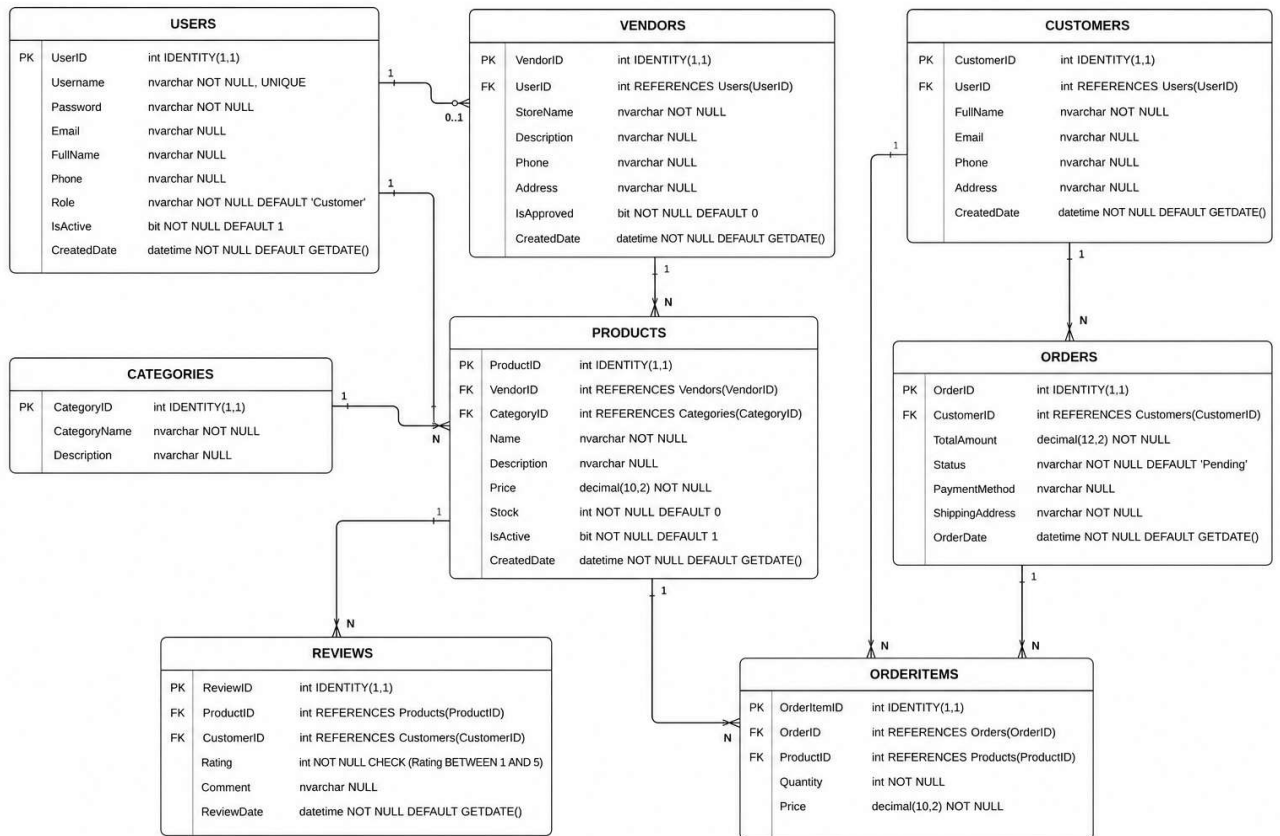
- **Store Dashboard:** As a vendor, I want to view analytics (total products, orders, revenue, and reviews) on my dashboard so that I can monitor my store's performance.
- **Inventory Control:** As a vendor, I want to add, update, and delete my products so that my active inventory remains accurate and up to date.
- **Fulfillment Tracking:** As a vendor, I want to track customer orders containing my products and update their fulfillment status (e.g., Processing, Shipped, Delivered) so that buyers know their order progress.
- **Sales Reporting:** As a vendor, I want to run visual sales reports of my top-performing products by revenue and quantity so that I can identify sales trends.

- Store Customization: As a vendor, I want to modify my store details (e.g., store name, phone, address, description) so that my business details look professional to buyers.

D. Administrator User Stories

- Central Dashboard Overview: As an admin, I want to monitor platform metrics (total registered accounts, approved vendors, active items, complete orders, total revenue) so that I can assess the overall health of the marketplace.
- User Oversight: As an admin, I want to view, search, filter, and deactivate user profiles (customers, vendors, and admins) so that I can maintain platform integrity.
- Vendor Approval Board: As an admin, I want to approve or reject newly registered vendors so that only verified merchants can list products on the store.
- Category Management: As an admin, I want to create, modify, and delete product categories so that the marketplace catalog remains organized.
- Review Moderation: As an admin, I want to view and delete product reviews that violate guidelines so that the review system remains constructive and trustworthy.

Chapter 3: Schema Diagram



Chapter 4: SQL Queries

```
CREATE DATABASE MultiVendorDB;  
GO  
USE MultiVendorDB;  
GO
```

```
CREATE TABLE Users (  
    UserID    INT PRIMARY KEY IDENTITY (1,1),  
    Username  NVARCHAR (50) NOT NULL UNIQUE,  
    Password  NVARCHAR (100) NOT NULL,  
    Email     NVARCHAR (100),  
    FullName  NVARCHAR (100),  
    Phone     NVARCHAR (20),  
    Role      NVARCHAR (20) NOT NULL DEFAULT 'Customer',  
    IsActive  BIT          NOT NULL DEFAULT 1,  
    CreatedDate DATETIME   DEFAULT GETDATE ()  
);
```

```
CREATE TABLE Vendors (  
    VendorID  INT PRIMARY KEY IDENTITY (1,1),  
    UserID    INT FOREIGN KEY REFERENCES Users (UserID),  
    StoreName NVARCHAR (100),  
    Description NVARCHAR (300),  
    Phone     NVARCHAR (20),  
    Address   NVARCHAR (200),  
    IsApproved BIT DEFAULT 0,  
    CreatedDate DATETIME DEFAULT GETDATE ()  
);
```

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY IDENTITY (1,1),  
    UserID     INT FOREIGN KEY REFERENCES Users (UserID),  
    FullName   NVARCHAR (100),  
    Email      NVARCHAR (100),  
    Phone      NVARCHAR (20),  
    Address    NVARCHAR (200),  
    CreatedDate DATETIME DEFAULT GETDATE ()  
);
```

```
CREATE TABLE Categories (  
    CategoryID INT PRIMARY KEY IDENTITY (1,1),  
    CategoryName NVARCHAR (100) NOT NULL,  
    Description NVARCHAR (200)  
);
```

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY IDENTITY (1,1),  
    VendorID  INT FOREIGN KEY REFERENCES Vendors (VendorID),
```



```

CategoryID INT FOREIGN KEY REFERENCES Categories (CategoryID),
Name NVARCHAR (150) NOT NULL,
Description NVARCHAR (500),
Price DECIMAL (10,2) NOT NULL,
Stock INT NOT NULL DEFAULT 0,
IsActive BIT NOT NULL DEFAULT 1,
CreateDate DATETIME DEFAULT GETDATE ()
);

```

```

CREATE TABLE Orders (
OrderID INT PRIMARY KEY IDENTITY (1,1),
CustomerID INT FOREIGN KEY REFERENCES Customers (CustomerID),
TotalAmount DECIMAL (12,2),
Status NVARCHAR (30) DEFAULT 'Pending',
PaymentMethod NVARCHAR (50),
ShippingAddress NVARCHAR (300),
OrderDate DATETIME DEFAULT GETDATE ()
);

```

```

CREATE TABLE OrderItems (
OrderItemID INT PRIMARY KEY IDENTITY (1,1),
OrderID INT FOREIGN KEY REFERENCES Orders (OrderID),
ProductID INT FOREIGN KEY REFERENCES Products (ProductID),
Quantity INT NOT NULL,
Price DECIMAL (10,2) NOT NULL
);

```

```

CREATE TABLE Reviews (
ReviewID INT PRIMARY KEY IDENTITY (1,1),
ProductID INT FOREIGN KEY REFERENCES Products (ProductID),
CustomerID INT FOREIGN KEY REFERENCES Customers (CustomerID),
Rating INT NOT NULL CHECK (Rating BETWEEN 1 AND 5),
Comment NVARCHAR (500),
ReviewDate DATETIME DEFAULT GETDATE ()
);

```

-- Default Users

```

INSERT INTO Users (Username,Password,Email,FullName,Role,IsActive)
VALUES ('admin','admin123', 'admin@shop.com','Administrator','Admin',1);

```

```

INSERT INTO Users (Username,Password,Email,FullName,Role,IsActive)
VALUES ('vendor','vendor123', 'vendor@shop.com','Tech Vendor','Vendor',1);

```

```

INSERT INTO Users (Username,Password,Email,FullName,Role,IsActive)
VALUES ('customer','cust123', 'customer@shop.com','John Customer','Customer',1);

```

```

INSERT INTO Vendors (UserID,StoreName,Description,Phone,Address,IsApproved)
VALUES (2,'TechZone Store','Best electronics!','9876543210','123 Tech Street',1);

```

```

INSERT INTO Customers (UserID,FullName,Email,Phone,Address)

```

VALUES (3,'John Customer', 'customer@shop.com','9123456789','456 Customer Lane');

INSERT INTO Categories (CategoryName,Description) VALUES ('Electronics','Gadgets');

INSERT INTO Categories (CategoryName,Description) VALUES ('Clothing','Fashion');

INSERT INTO Categories (CategoryName,Description) VALUES ('Books','Education');

INSERT INTO Categories (CategoryName,Description) VALUES ('Home & Kitchen','Appliances');

INSERT INTO Categories (CategoryName,Description) VALUES ('Sports','Equipment');

INSERT INTO Products

(VendorID,CategoryID,Name,Description,Price,Stock,IsActive)

VALUES (1,1,'Wireless Headphones','Bluetooth headphones',2499.00,50,1);

INSERT INTO Products

(VendorID,CategoryID,Name,Description,Price,Stock,IsActive)

VALUES (1,1,'Smartphone X200','6.5-inch AMOLED',18999.00,30,1);

INSERT INTO Products

(VendorID,CategoryID,Name,Description,Price,Stock,IsActive)

VALUES (1,2,'Cotton T-Shirt','Premium cotton',499.00,200,1);

INSERT INTO Products

(VendorID,CategoryID,Name,Description,Price,Stock,IsActive)

VALUES (1,3,'Python Book','Complete Python guide',449.00,60,1);

INSERT INTO Products

(VendorID,CategoryID,Name,Description,Price,Stock,IsActive)

VALUES (1,4,'Electric Kettle','1.5L kettle',1199.00,45,1);

GO

Chapter 5: Screenshots

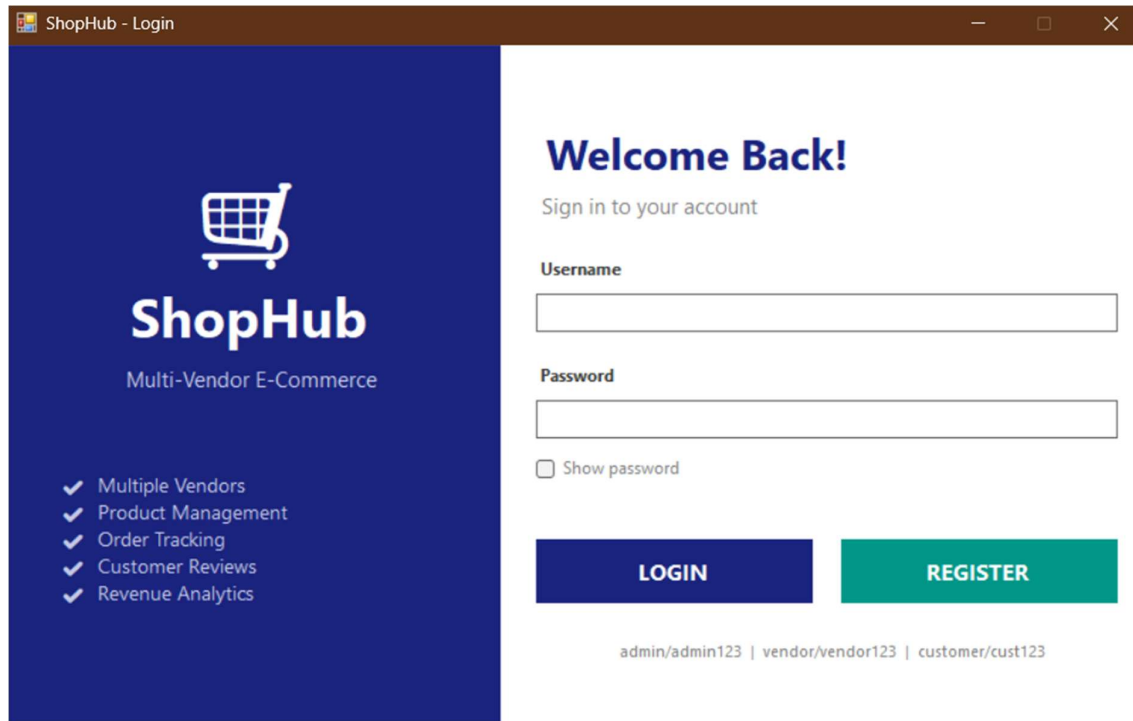


Figure 01: Login/Registration Panel

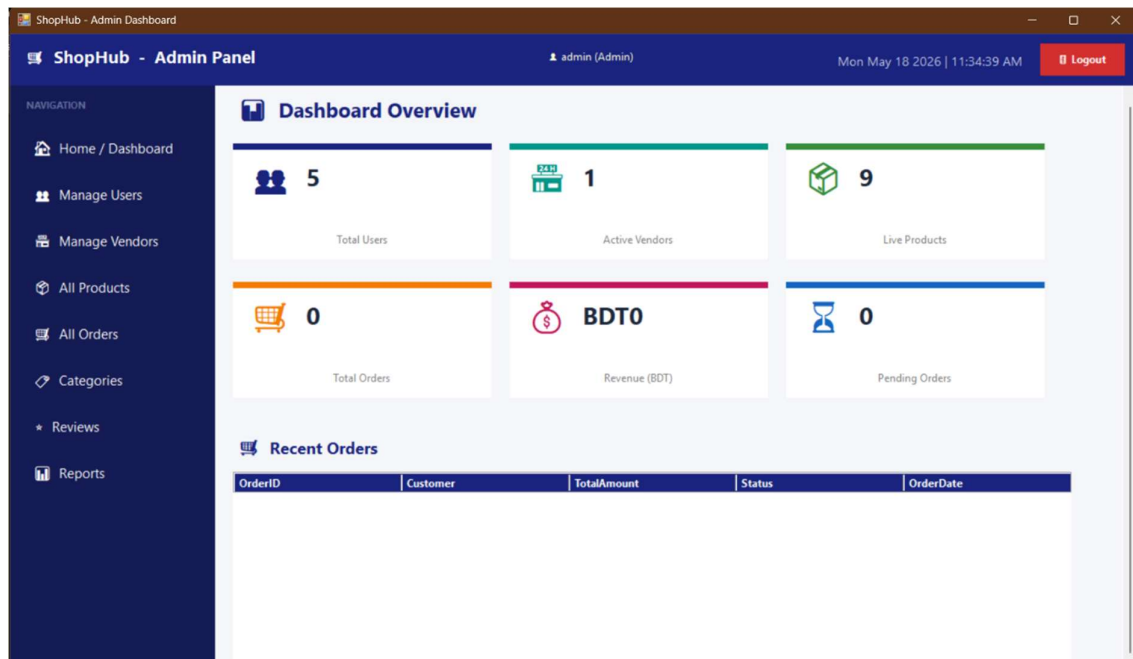


Figure 02: Admin Dashboard

Manage Users

Manage Users

Search:

Role:

All

Toggle Active

Delete

UserID	Username	FullName	Email	Phone	Role	Status	Joined
5	Sakib	sakib	s@gmail.com	01771259004	Customer	Active	18/05/2026
2	admin	Administrator	admin@shop.com		Admin	Active	17/05/2026
3	vendor	Tech Vendor	vendor@shop.com		Vendor	Active	17/05/2026
4	customer	John Customer	customer@shop.com		Customer	Active	17/05/2026
1	superadmin	System Controller	sa@bd.com		SuperAdmin	Active	17/05/2026

Manage Vendors

Manage Vendors

✓ Approve

✗ Reject

Refresh

VendorID	Username	StoreName	Phone	Address	Status	Joined
1	admin	TechZone Store	9876543210	123 Tech Street	Approved	17/05/2026

All Products

All Products

Search:

Toggle Active

ProductID	Name	Category	Vendor	Price	Stock	Status
9	Dell Laptop	Books	TechZone Store	45000.00	15	Active
8	Toyota Corolla	Clothing	TechZone Store	3500000.00	2	Active
7	MacBook Air M2	Electronics	TechZone Store	110000.00	5	Active
6	iPhone 14 Pro	Electronics	TechZone Store	95000.00	10	Active
5	Electric Kettle	Home & Kitchen	TechZone Store	1199.00	45	Active
4	Python Book	Books	TechZone Store	449.00	60	Active
3	Cotton T-Shirt	Clothing	TechZone Store	499.00	200	Active
2	Smartphone X200	Electronics	TechZone Store	18999.00	30	Active
1	Wireless Headphones	Electronics	TechZone Store	2499.00	50	Active

All Orders

All Orders

Set Status:

Apply

Refresh

OrderID	Customer	TotalAmount	Status	PaymentMethod	OrderDate	ShippingAddress
---------	----------	-------------	--------	---------------	-----------	-----------------

Manage Categories

List

+ Add / Edit

CategoryID	CategoryName	Description
3	Books	Education
2	Clothing	Fashion
1	Electronics	Gadgets
4	Home & Kitchen	Appliances
5	Sports	Equipment

Manage Reviews

★ Manage Reviews

Delete Review

Refresh

ReviewID	Product	Customer	Rating	Comment	ReviewDate
----------	---------	----------	--------	---------	------------

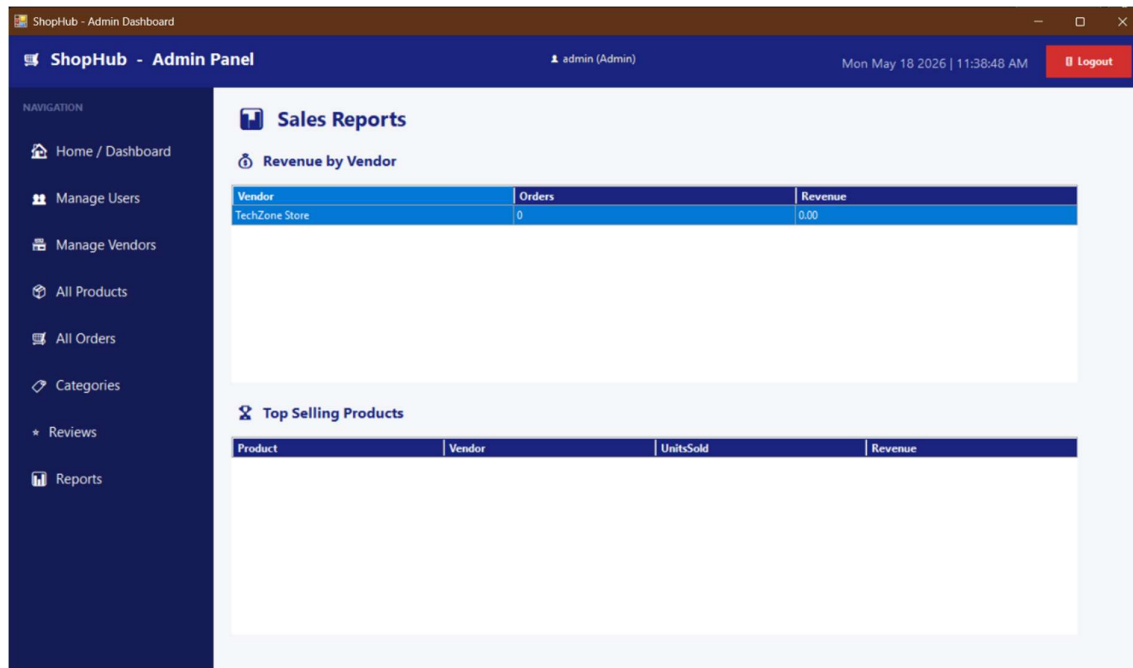


Figure 03: Admin Controls

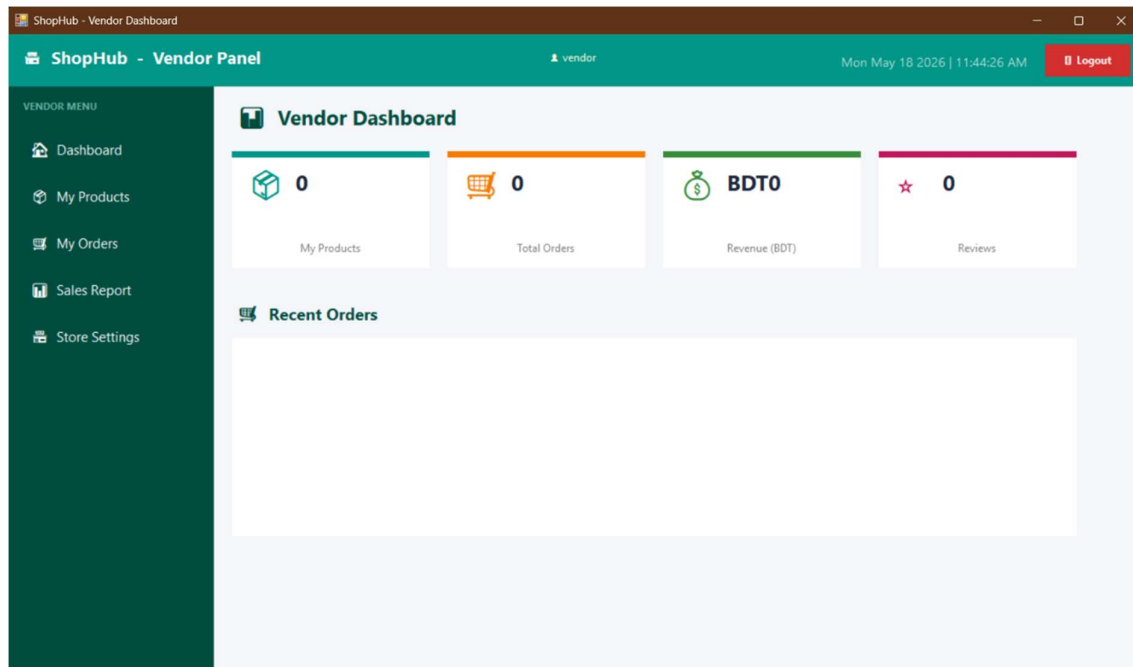


Figure 04: Vendor Dashboard

My Products

My Products

+ Add / Edit

My Products

Refresh

ProductID	Name	Category	Price	Stock	Status	Description
-----------	------	----------	-------	-------	--------	-------------

My Orders

My Orders

Status: Pending Update

The image shows two overlapping browser windows from a web application. The top window is titled 'ShopHub - Vendor Dashboard' and displays a 'Sales Report' section with a table header. The bottom window is titled 'Store Settings' and contains a form with four input fields.

ShopHub - Vendor Dashboard

Vendor Panel | vendor | Mon May 18 2026 | 11:45:07 AM | Logout

VENDOR MENU

- Dashboard
- My Products
- My Orders
- Sales Report
- Store Settings

Sales Report

Top Products by Revenue

Product	UnitsSold	Revenue
---------	-----------	---------

Store Settings

Store Name

Phone

Address

Description

Figure 05: Vendor Controls

ShopHub - Customer Store

ShopHub - Online Store customer Mon May 18 2026 | 11:50:28 AM Logout

CUSTOMER MENU

- Shop / Browse
- My Cart
- My Orders
- Write Review
- My Profile

Browse Products

All Categories Search

ProductID	Name	Category	Vendor	Price	Stock	Rating
9	Dell Laptop	Books	TechZone Store	45000.00	15	0
8	Toyota Corolla	Clothing	TechZone Store	3500000.00	2	0
7	MacBook Air M2	Electronics	TechZone Store	110000.00	5	0
6	iPhone 14 Pro	Electronics	TechZone Store	95000.00	10	0
5	Electric Kettle	Home & Kitchen	TechZone Store	1199.00	45	0
4	Python Book	Books	TechZone Store	449.00	60	0
3	Cotton T-Shirt	Clothing	TechZone Store	499.00	200	0
2	Smartphone X200	Electronics	TechZone Store	18999.00	30	0
1	Wireless Headphones	Electronics	TechZone Store	2499.00	50	0

Add to Cart

Figure 06: Customer Dashboard

ShopHub - Customer Store

ShopHub - Online Store customer Mon May 18 2026 | 11:50:44 AM Logout

CUSTOMER MENU

- Shop / Browse
- My Cart
- My Orders
- Write Review
- My Profile

My Cart

ProductID	Name	Price	Quantity	SubTotal

Total: BDT0.00

Cash on Delivery

Remove Place Order Clear Cart

My Orders

My Orders

OrderID	TotalAmount	Status	PaymentMethod	ShippingAddress	OrderDate
---------	-------------	--------	---------------	-----------------	-----------

Write a Review

★ Write a Product Review

Select Product

Rating (1-5 Stars)

★★★★★ (5)

Your Comment

★ Submit Review

 My Profile

 **My Profile**

Full Name

Sakib

Email

sakib@gmail.com

Phone

01771259004

Address

uttara, dhaka

 Save Profile

Change Password

Old Password

New Password

 Change Password

Figure 07: Customer Access